Daniel Liu

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About Me

Professional Interests: Machine learning, data science, engineering, and visualization, alpha research

Languages: Python, Java, C++, TypeScript, Matlab, C (GCC/clang), Fortran (f95)

Technologies: Git, Unix, Shell, AWS, Jupyter/pandas, matplotlib, XGBoost, LightGBM, PostgreSQL, Serpent, MCNP, LAMMPS

Legal Status: Canadian Citizen. Available for H1B or TN visa.

Experience

Terrestrial Energy
Toronto

Reactor Physicist & Software Engineering
January 2020 – Present

- Developed core physics parameters for a next-generation molten salt reactor using monte carlo codes (Serpent & MCNP)
- Investigating CPU/GPU offloading techniques to speed-up radiation transport simulations using OpenACC / ROCm
- Extended high performance concurrent C++ utilities to couple rapidly updating physical data sets for nuclear simulation
- Developing a full-scope simulator by integrating mathematical first principles on nuclear phenomena (i.e., thermo-hydraulics, radiation shielding design, nuclear fuel burnup)

Ontario Power Generation
Toronto

Software Engineer - Nuclear Simulations
September 2018 – December 2019

- Developed a heavy water decontamination scheduler for all nuclear reactors in Canada using VBA & Python
- Designed the configuration of radiation shielding using monte carlo analysis on various gamma-ray emitting sources
- Created a real time concurrent C++ service that tracks and publishes logic decisions for plant control
- Optimized SQL database queries on multiple ODBC connections to push maintenance notices on critical nuclear equipment

Canadian Nuclear Laboratories
Chalk River

Software Engineering R&D - Nuclear Waste Simulations
May 2018 – September 2018

- Developed code to model the flow of groundwater surrounding below-grade nuclear waste repositories
- Designed additional modules describing the interaction of dissolved chemical species with the surrounding earth
- Provided detailed project updates to federally run Atomic Energy of Canada Limited (AECL) to maintain funding of department Science & Technology experiments worth >$ 200 k for further development and eventual commercialization
- Developed matlab code to simulate gas-catalyst interactions to develop and scale-up an experimental process to separate Tritium from other gasses, a radioactive isotope of Hydrogen used for fusion energy research.

Projects and Activities

10-Q/10-K Insights : Created a scraper using Beautiful Soup in Python to aquire market quarterly and annual earning reports to assess for predictive relationships between qualitative risk factors and future returns

McKinsey & Company Hackathon : Analyzed if a requested loan is approved or not, based on the features of the applicant (e.g., income, gender, city), and that of the loan (e.g., loan amounts, interest rates). Used XGBoost / gradient boosting for analysis

Scotiabank Hackathon : Concepted an addicting mobile game to help reduce user debt with in-app purchases routed back to saving accounts. Participated with seven software engineers and graphics designers within 48 hours to win a $10,000 prize

Nuclear Thermal Propulsion: Preparing a business case study on the advantages of developing nuclear thermal propulsion technologies in Canada. Leading the development of physics and engineering proposals with nuclear laboratories in North America

Hackintosh: Modified MacOS Sierra kernel extension files to enable PC compatibility with a GTX 1070

Education

University of Ontario Institute of Technology *2013 – 2018*

Bachelor of Engineering – Nuclear Engineering, Minors in Computer Science (95% cGPA)

Research : Created a MATLAB application to track AR device user location in GPS denied environments using MEMS input sensor data.